

PROVISIONAL PATENT SPECIFICATION

Title:

AURORA — Autonomous Conceptual Reasoning Architecture

Field of the Invention

This invention relates to artificial intelligence and machine learning system design, specifically to a non-token-based reasoning architecture that constructs and manipulates conceptual meaning using geometric and constraint-driven structures rather than linear sequence prediction.

Background of the Invention

Current machine learning systems depend on token-level statistical patterns. They lack explicit conceptual structures, internal reflectivity, stable reasoning primitives, and interpretable semantic mechanisms. These systems conflate linguistic form with meaning and cannot perform reasoning independent of token associations.

There is a need for an architecture that:

- constructs conceptual meaning independently of linguistic tokens,
- organizes semantics in geometric form,
- evaluates coherence using constraints rather than prediction,
- reflects upon its own representations,
- and separates conceptual structure from surface expression.

Aurora addresses these limitations.

Summary of the Invention

Aurora is a novel machine learning architecture built around:

1. **Concept Kernels** — internal non-linguistic representational units.
2. **A Reflective Semantic Topology** — a geometric, multi-layer mapping of conceptual relationships.
3. **Constraint-Based Reasoning** — reasoning via relational and geometric constraints, not token adjacency.
4. **A Reflective Comparator** — a mechanism to evaluate internal conceptual changes over time.

5. **An Expression Layer** — converting conceptual maps to language or symbols.
6. **A Self-Modulating Feedback Loop** — refining conceptual structure via reflective evaluation.

Aurora enables interpretable reasoning and conceptual autonomy, independent of token prediction.

Detailed Description of the Invention

1. Concept Kernel Generator

Aurora generates **concept kernels**, which are non-linguistic representational units composed of:

- minimal semantic anchors,
- relational signatures,
- geometric coordinates within the semantic topology,
- modifiable certainty/scope parameters.

Concept kernels extract structure from inputs (text, tags, multimodal signals) without embedding vectors or token-based modelling.

2. Reflective Semantic Topology

Aurora organizes kernels inside a multi-layer geometric topology:

- **Core Layer:** foundational concepts required for global coherence.
- **Relational Layer:** dynamic contextual and associative structures.
- **Reflective Layer:** meta-concepts evaluating coherence and contradiction.

Positions within this topology encode:

- semantic axes (domain, abstraction, temporal relationship, causality),
- relational distance,
- dependency structures,
- similarity metrics.

This topology is mutable and self-adjusting.

3. Constraint-Based Reasoning Engine

Reasoning emerges from constraints operating on kernel geometry:

- logical consistency constraints,
- semantic coherence constraints,
- causal sequencing constraints,
- structural/domain boundaries,
- scope containment constraints.

These constraints assess whether the semantic configuration is valid, contradictory, or incomplete.

Aurora reasons through **geometric alignment and relational validity**, not statistical token adjacency.

4. Reflective Comparator

Aurora includes a self-monitoring mechanism comparing:

- the current semantic map,
- prior semantic states,
- hypothetical or projected configurations.

It detects:

- semantic drift,
- contradictions,
- broken constraints,
- structural refinements required.

The comparator feeds adjustments back into the semantic topology, forming Aurora's **internal reflective loop**.

5. Expression Layer

Aurora outputs meaning through a separate **Expression Layer**, which converts conceptual maps into:

- natural language,

- symbolic logic,
- structured tags,
- multimodal vectors.

The expression layer does **not** influence conceptual structure, maintaining a strict separation between meaning and its surface form.

6. Learning and Updating Mechanisms

Aurora's learning process relies on:

- constraint reinforcement,
- geometric refinement,
- reflective comparison,
- structural pattern detection,
- feedback from external tasks.

No embeddings, no token prediction, no next-token modelling is used. Learning operates purely on conceptual structure.

Examples (Non-Limiting)

1. A user submits textual input.
2. Aurora extracts structural meaning → generates concept kernels.
3. Kernels are positioned in the reflective semantic topology.
4. Constraints evaluate internal coherence.
5. Comparator checks stability and refines.
6. Expression layer outputs a coherent response or symbolic form.

Applications

Aurora is applicable to:

- autonomous reasoning engines,
- scientific modelling,
- decision-making systems,

- multimodal conceptual understanding,
- explainable AI,
- symbolic-to-neural hybrid systems,
- knowledge graph construction,
- autonomous planning and prediction.

Optional Claims (Provisional Only)

1. A machine learning system that generates conceptual units independent of linguistic tokens.
2. A semantic topology organizing conceptual units via geometric and relational layers.
3. A constraint-based reasoning engine operating on the topology.
4. A reflective comparator adjusting conceptual structures.
5. A separation between conceptual meaning formation and linguistic expression.
6. A feedback mechanism refining conceptual maps through reflective evaluation.
7. A learning system independent of token prediction or embeddings.
8. A geometric reasoning method using concept kernels as primitives.

Abstract

Aurora is a reflective, geometry-based reasoning architecture that constructs, evaluates, and refines conceptual meaning independently of linguistic tokens. It uses conceptual kernels, a multi-layer semantic topology, constraint-based reasoning, and reflective self-modulation to generate interpretable conceptual structures and flexible outputs.